

```

5 def filter_data(
6     df: pl.DataFrame,
7 ) -> pl.DataFrame:
8
9     df = df.filter(
10         (pl.col("defense_system_cluster_id") == pl.col("cluster_id")) &
11         (pl.col("Organism") == "Homo sapiens (Human)")
12     )
13     logger.info(f"N rows after filtering: {len(df)}")
14
15     return df

```

```

17 # What are the species annotations for the filtered data?
18 print("Unique values for")
19 print(df_def_cleaned["Organism"])
20
21 df_def_filtered = filter_data(
22     df=df_def_cleaned,
23 )
24
25 df_def_filtered.write_csv(
26     f"{args['out_path']}.csv",
27     separator="\t",
28     include_header=True
29 )
30

```

```

TCAG
TCAGT
TCAGTC
GTCAGTCA
TCAGTCA
CAGTCAG
AGTCAG
TCAGTC
TCAGTC
CAGTCAG
AGTCAGTC
TCAGTCAGTCAGTCAG
GTCAGTCAGTCAG
GTCAGTCAG
GTCA TCAGT
AGTCA CAGTCA
CAGTCAG AGTCAGTC
CAGTCAGT TCAGTCA
AGTCAGT TCAGTCAG
AGTCAGTC CAGTCAGT
GTCAGTCA GTCAGTCA
GTCAGTCAG TCAGTCAG
CAGTCAG AGTCAGT
CAGTCAGT GTCAGTC
AGTCAGT CAGTCAG
TCAGTCA CAGTCAGT
CAGTCAG AGTCAGTC
AGTCAG AGTCAGTC
GTCAGTC TCAGTCAG
TCAGTC CAGTCAGT
AGTCAG AGTCAGT
TCAG AGTCAGT

```

IEI project: update 2

Julian Moran

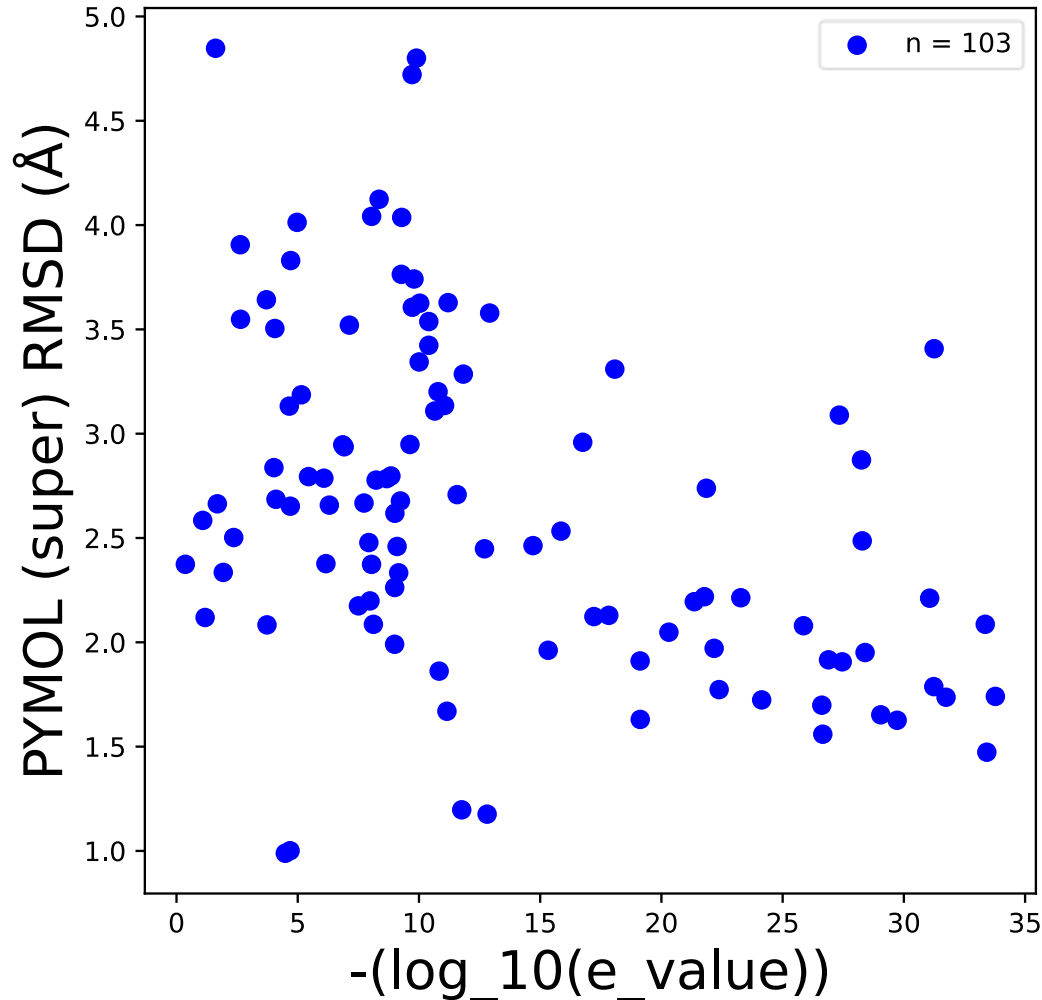
2026-04-10

The Centre for Applied Genomics, SickKids

Aims

1. Automate pairwise structural alignment with PyMOL
2. Manually curate positive control set of bacteria-human protein pairs
 - source: high-impact literature
 - purpose: to serve as positive control in subsequent analysis
 - criteria: bacterial homology to human, proof of bacterial immune function
3. Build pipeline frontend user interface for interactive analysis
 - target users: C Trost lab, J Moran, bacterial protein research community
 - features: structural alignment visualization, sequence alignment visualization, domain annotation overlay
4. Run pipeline on all bacterial defense systems
 - incurs prioritization problem
5. Get gene list
 - target users: C Marshall clinical research group
 - purpose: improve diagnostic yield for rare single-variant immune disease from genome sequencing

Why use PyMol to check structural alignment?



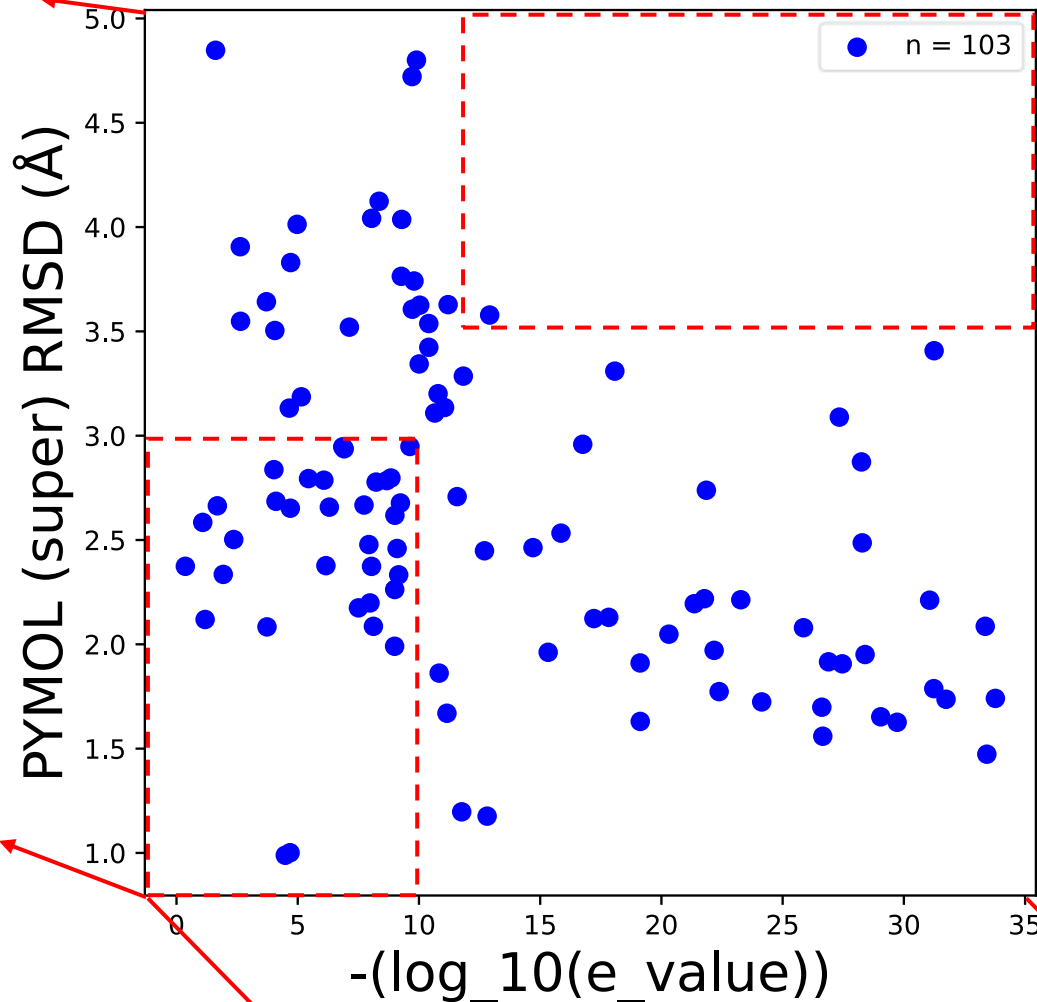
e-values [given by Foldseek] denote the probability of observing this level of structural alignment due to random chance

They are the principal measure of structural similarity in Foldseek

Why the discrepancy between e-value and PyMOL structural alignment?

Why use PyMol to check structural alignment?

Worse
structural
alignment
according
to PyMol



Better
structural
alignment
according
to PyMol

worse structural alignment
according to Foldseek

e-values [given by Foldseek] denote the probability of observing this level of structural alignment due to random chance

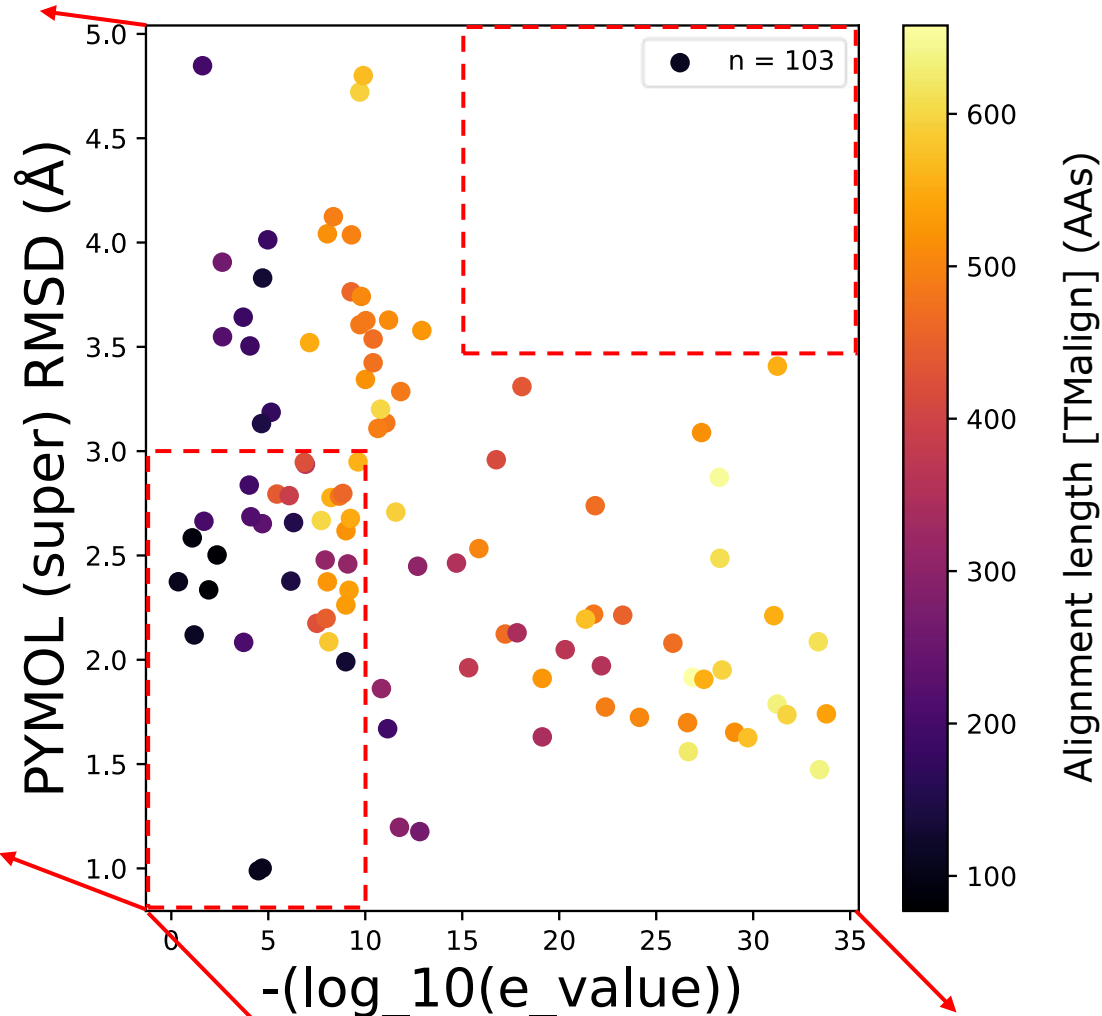
They are the principal measure of structural similarity in Foldseek

Why the discrepancy between e-value and PyMOL structural alignment?

Better structural alignment
according to Foldseek ($e = 10^{-35}$)

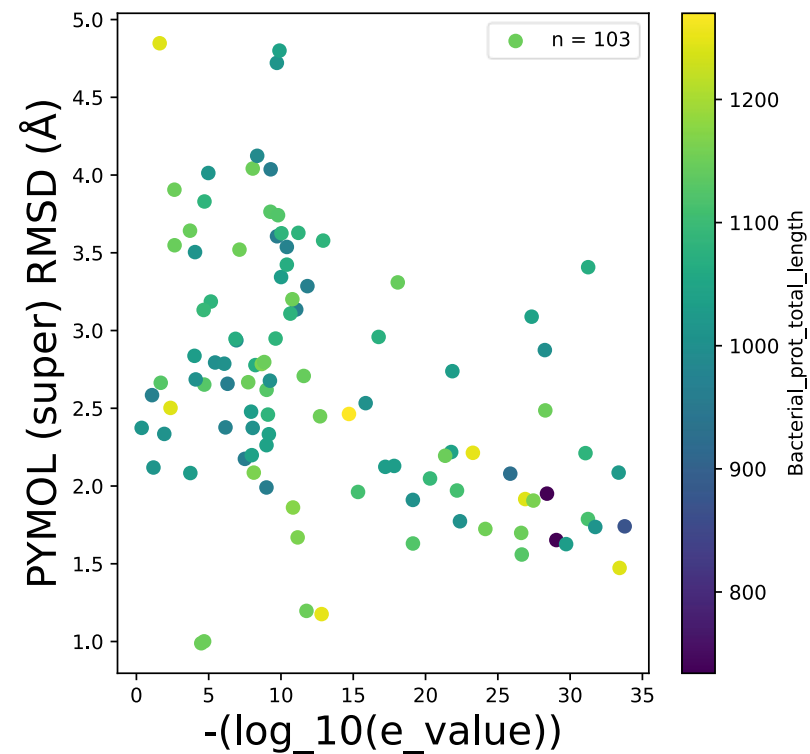
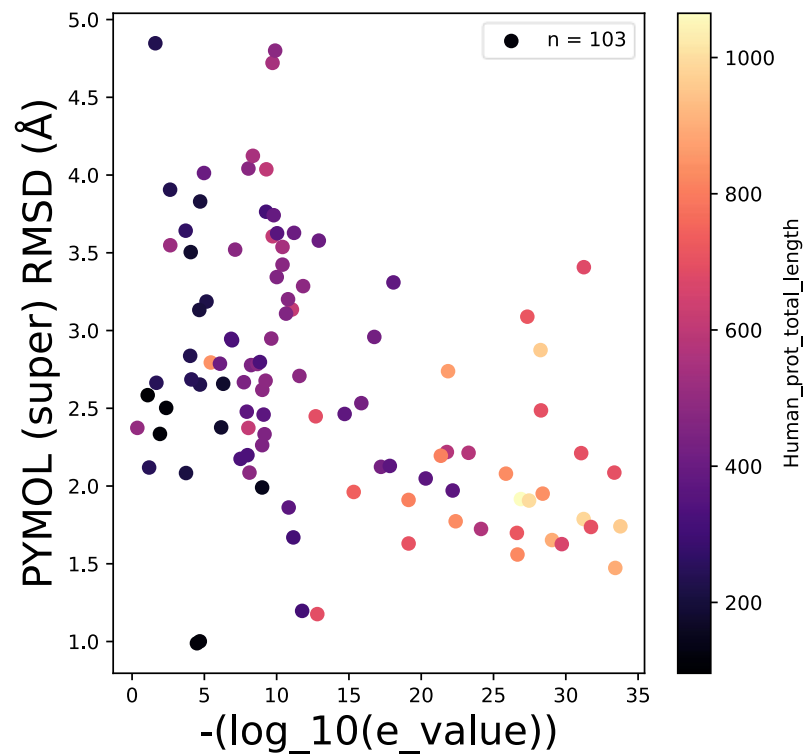
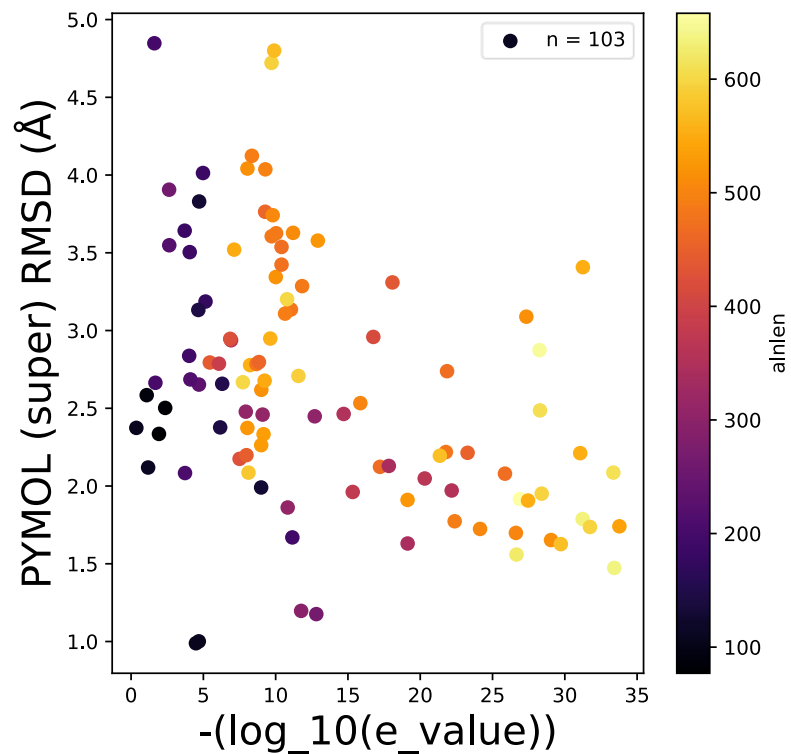
Why use PyMol to check structural alignment?

Worse
structural
alignment
according
to PyMol

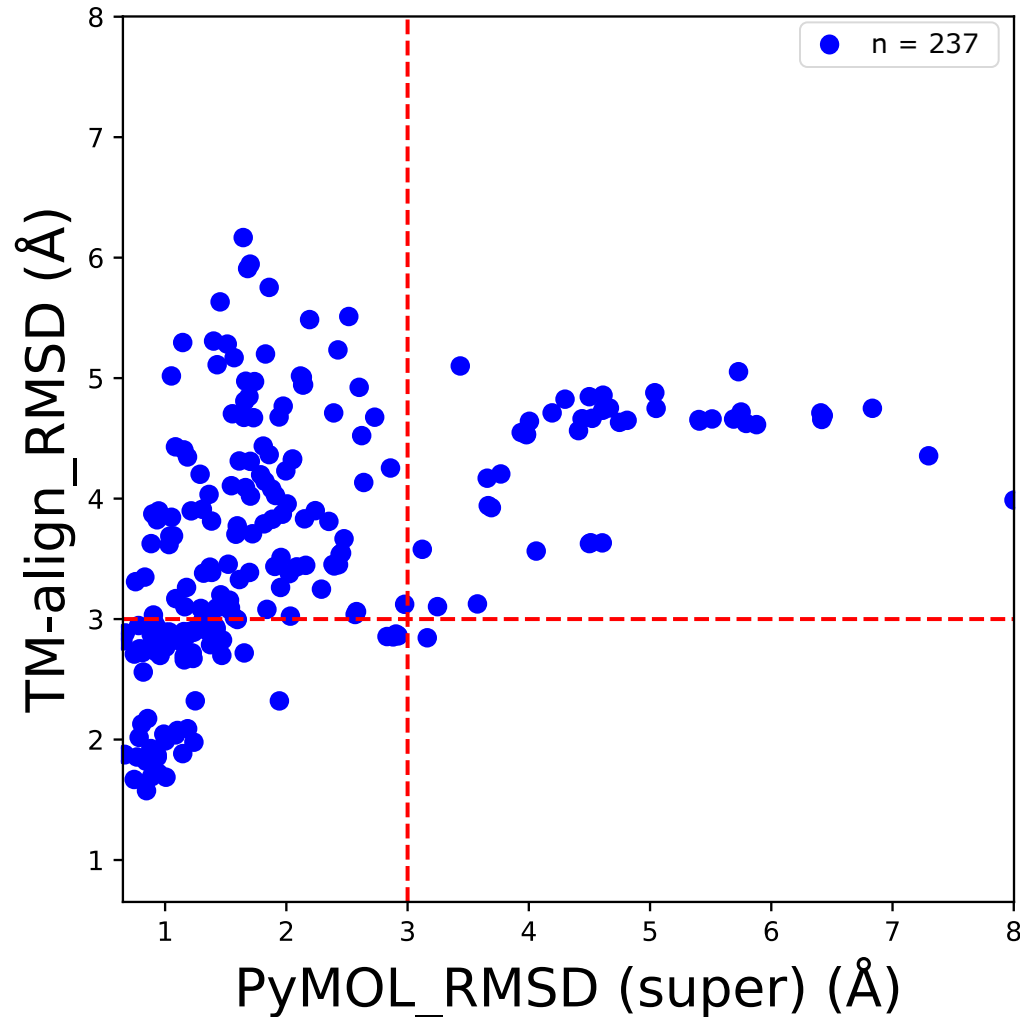


e-values are contingent on alignment length, which means filtering for only the most significant e-values risks missing domain-specific alignment

Conversely, PyMOL alignments can be good but insignificant



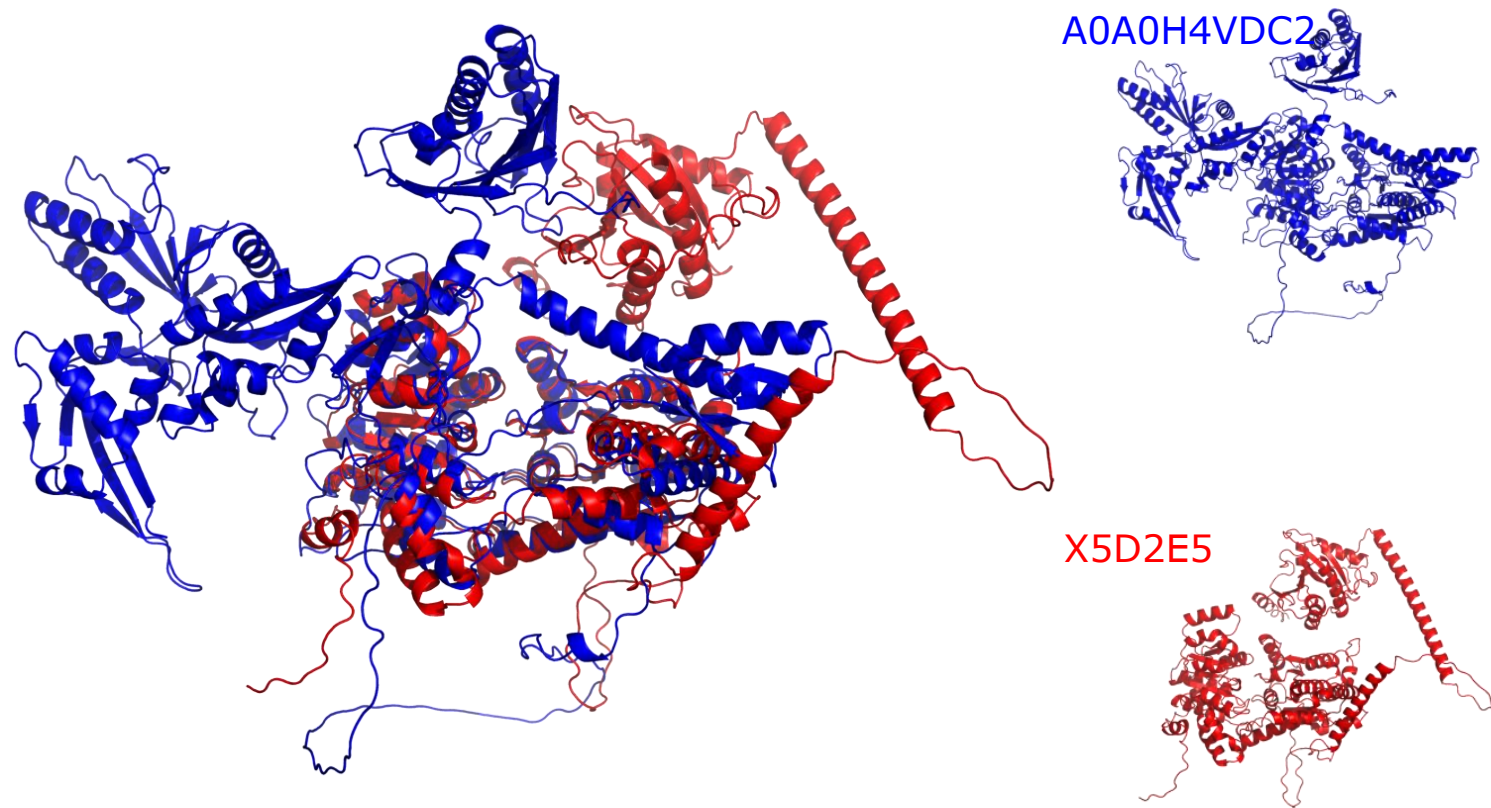
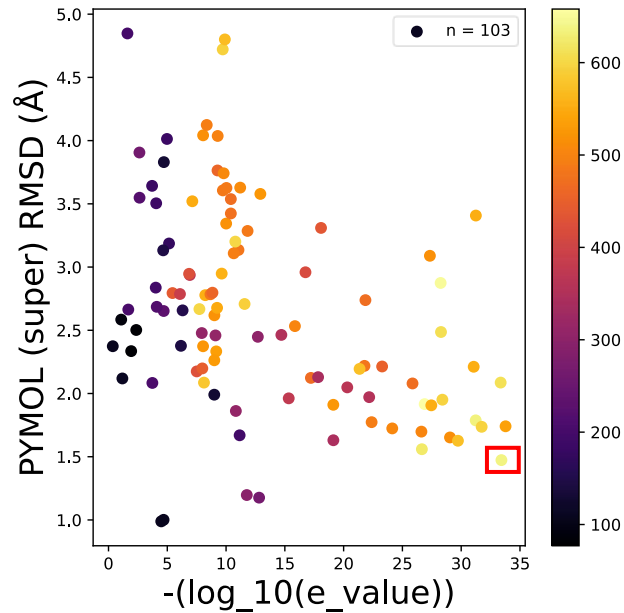
Why use PyMol to check structural alignment?



PyMol super can achieve good alignment when TM-align cannot (see top-left quadrant vs. bottom-right).

However, this should always be checked against a threshold of $e < 0.05$ (after multiple hypothesis correction) to confirm that the alignment is not due to random chance.

**To this end, we have
incorporated PyMol super into
our pipeline for interactive
filtering**



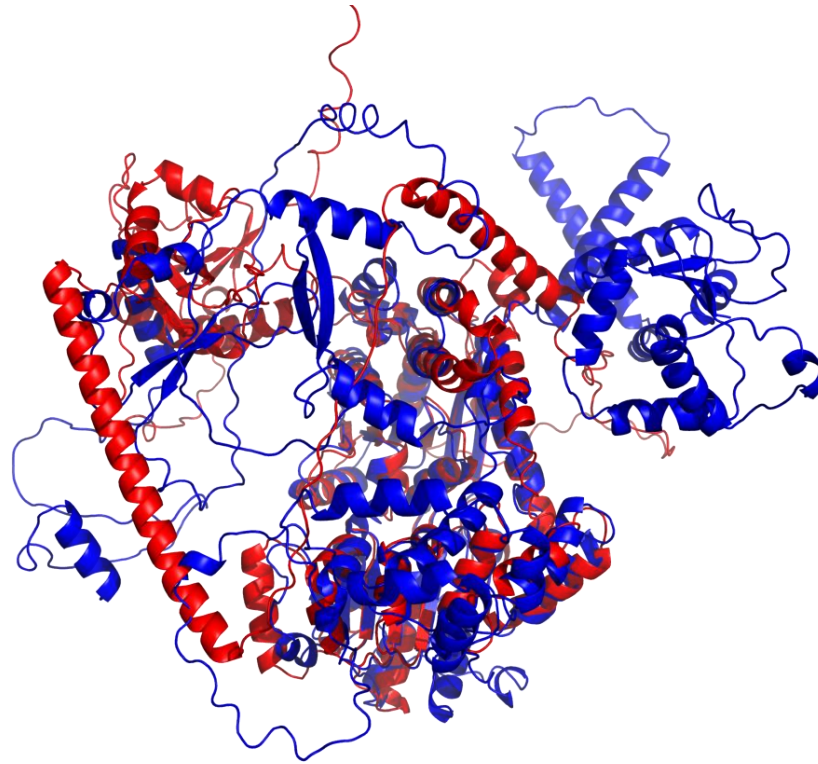
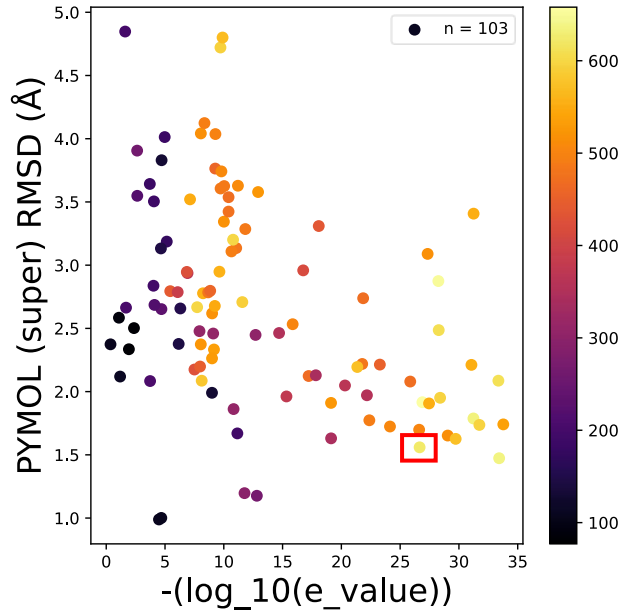
System	Zorya
e-value	$3.8e^{-34}$
PyMol RMSD	1.47 Å
Aln length	636 AAs

A0A0H4VDC2 [A. atlanticus]

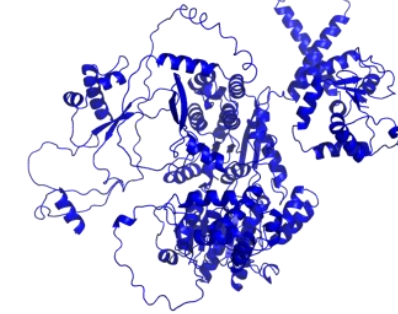
- unreviewed, uncharacterised protein
- strong sequence homology to ATP-, DEAD-box helicases in other bacterial species

X5D2E5 [H. sapiens][CHD1L]

- ATP-binding, helicase activity
- "binds histones that are poly-ADP-ribosylated on serine residues in response to DNA damage"



A0A5B8XTL3



A0A0A0MSH9



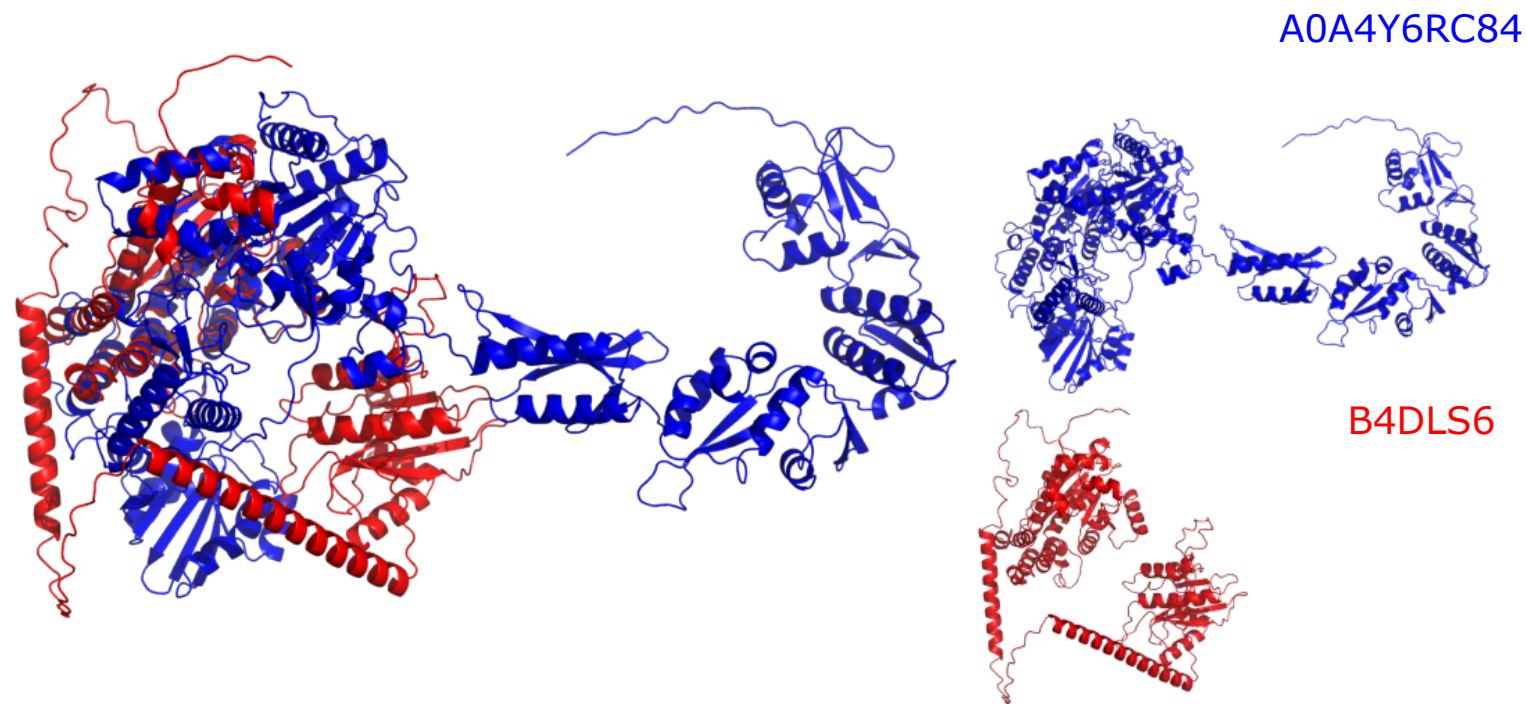
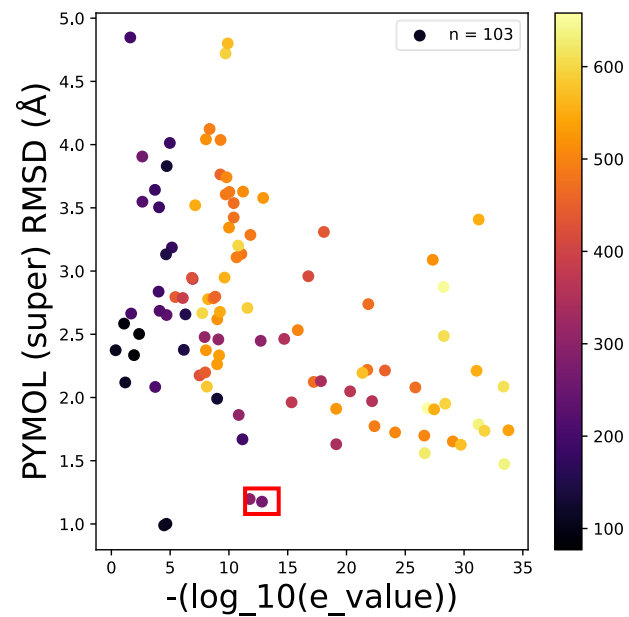
System	Zorya
e-value	$2.21e^{-27}$
PyMol RMSD	1.56 Å
Aln length	623 AAs

A0A5B8XTL3 [M. marinus]

- DEAD-box helicase

A0A0A0MSH9 [H. sapiens][CHD1L]

- ATP-binding, helicase activity
- "binds histones that are poly-ADP-ribosylated on serine residues in response to DNA damage"



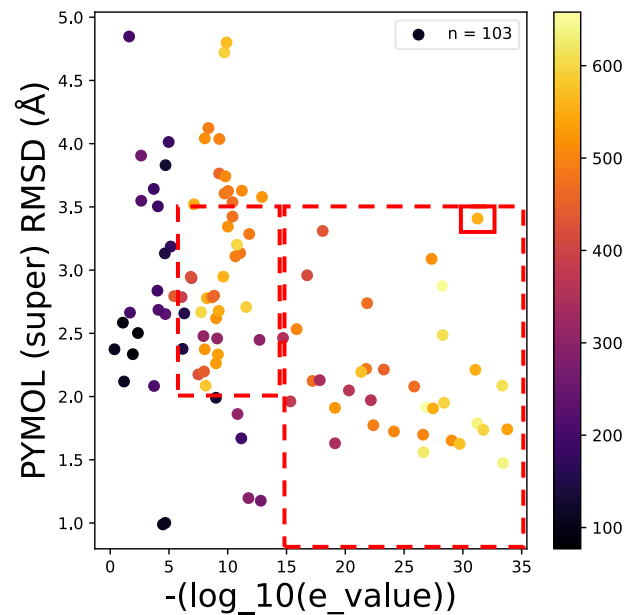
System	Zorya
e-value	$1.55e^{-13}$
PyMol RMSD	1.17 Å
Aln length	267 AAs

A0A4Y6RC84 [*J. tructae*]

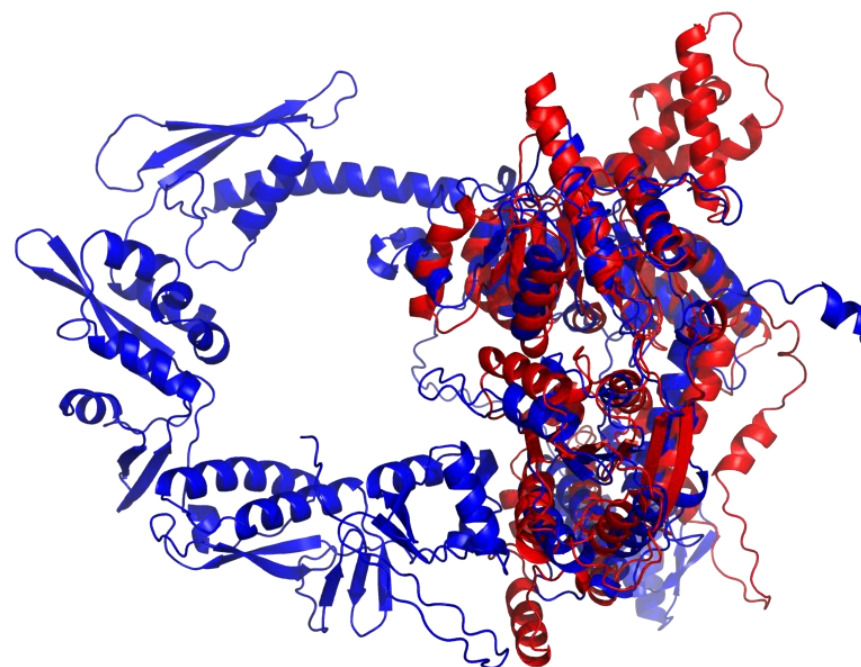
- ATP-dependent helicase
- unreviewed

B4DLS6 [*H. sapiens*][unreviewed]

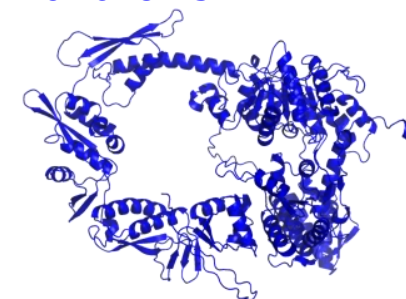
- "highly similar to Homo sapiens chromodomain helicase DNA binding protein 1-like (CHD1L)"



System	Zorya
e-value	$5.65e^{-32}$
PyMol RMSD	3.41 Å
Aln length	553 AAs



A0A6L8M512



Q8N6J1



A0A6L8M512 [A. baumannii]

- DEAD box helicase protein

Q8N6J1 [H. sapiens][BTAF1]

- evidence of existence only at transcript level
- predicted helicase, hydrolase activity

Curating a positive control set of homologous bacterial-human protein pairs from high-impact literature

Experimental criteria

provide protein accession for bacterial protein

express in new organism

perform functional assay proving bacterial immune system action

Homology criteria

provide protein accession for human protein

present convincing evidence of sequence or structure homology

Positive control set

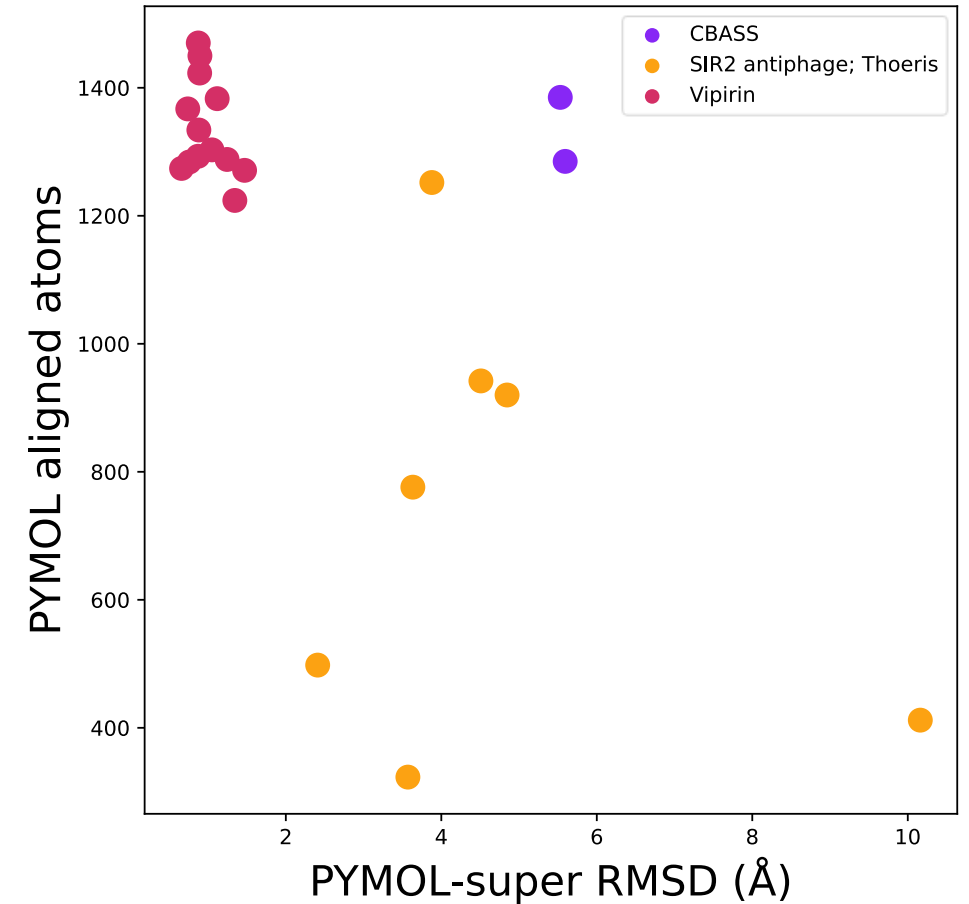
Source: high-impact literature

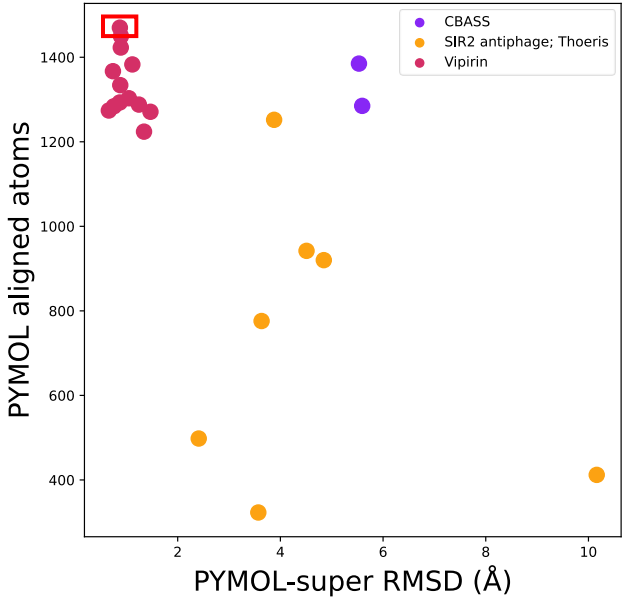
Criteria: bacterial homology to human, proof of bacterial immune function

Purpose: positive control for subsequence analysis

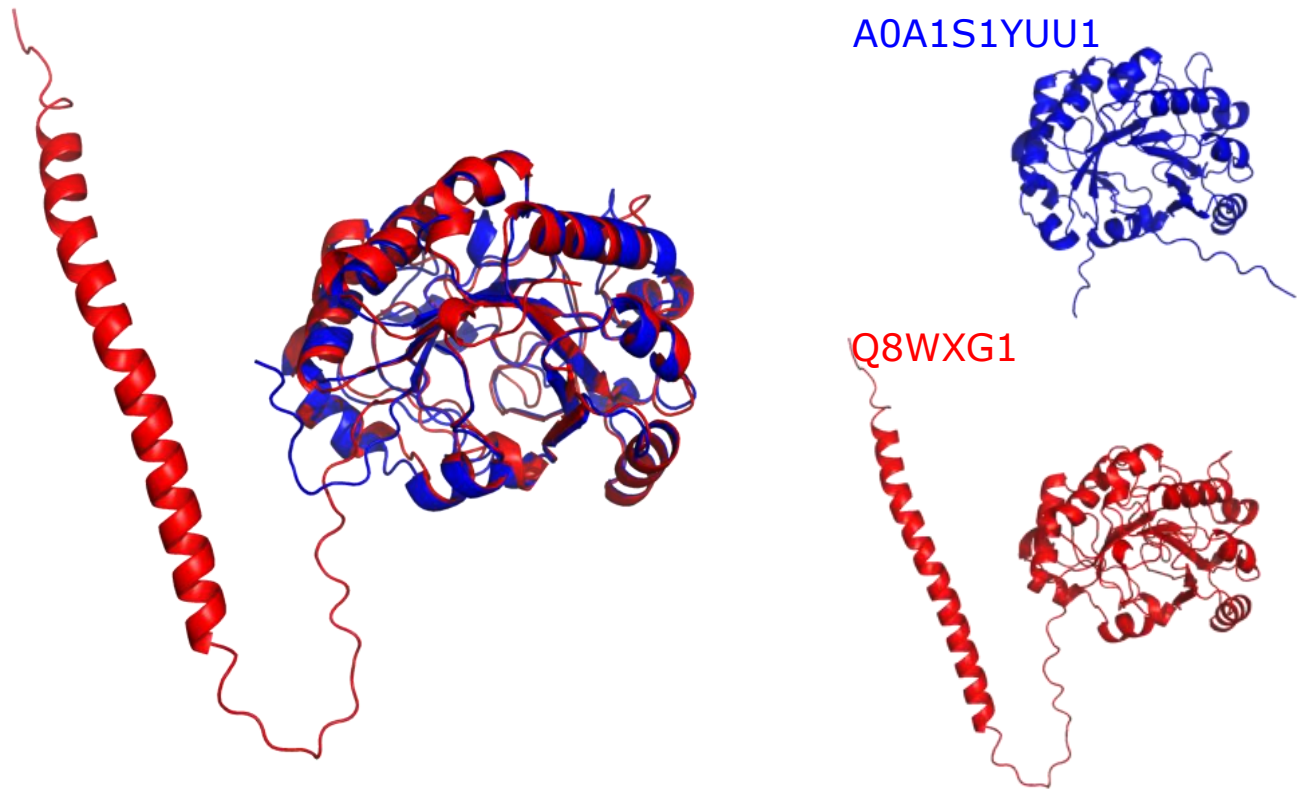
PMID	Bact. UniProt	Bact. gene name	H.s. UniProt	H.s. gene name
30787435	Q6XGD5	cdnE	Q8N884	CGAS
30787435	G2SLH8	cdnE	Q8N884	CGAS
40705877	J8G6Z1	ThsA	Q9BPY3	FAM118B
40705877	J8G6Z1	ThsA	Q96EB6	SIRT1
40705877	J8G6Z1	ThsA	Q8IXJ6	SIRT2
40705877	J8G6Z1	ThsA	Q9Y6E7	SIRT4
40705877	J8G6Z1	ThsA	Q9NXA8	SIRT5
40705877	J8G6Z1	ThsA	Q8N6T7	SIRT6
40705877	J8G6Z1	ThsA	Q9NRC8	SIRT7
32937646	A0A1H0NKS3	vip6	Q8WXG1	RSAD2
32937646	P0DW49	vip8	Q8WXG1	RSAD2
32937646	D5EID4	vip15	Q8WXG1	RSAD2
32937646	P0DW52	vip21	Q8WXG1	RSAD2
32937646	A0A1S1YUU1	vip47	Q8WXG1	RSAD2
32937646	A0A110A2W7	vip50	Q8WXG1	RSAD2
32937646	A0A1M7D0R2	vip56	Q8WXG1	RSAD2
32937646	A0A244CMP0	vip58	Q8WXG1	RSAD2
32937646	P0DW53	vip60	Q8WXG1	RSAD2

Total number: 18 pairs





System	Vipirin
PyMol RMSD	0.89 Å
Aln length	1423 atoms [75 AAs]

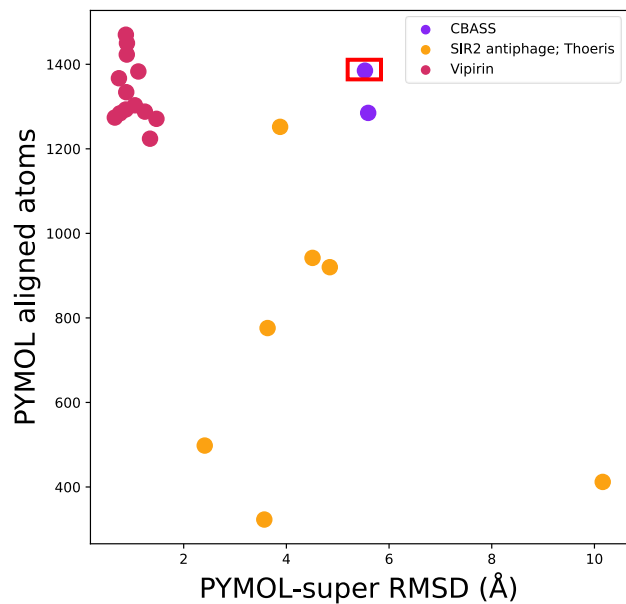


A0A1S1YUU1 [F. pacifica][vip47]

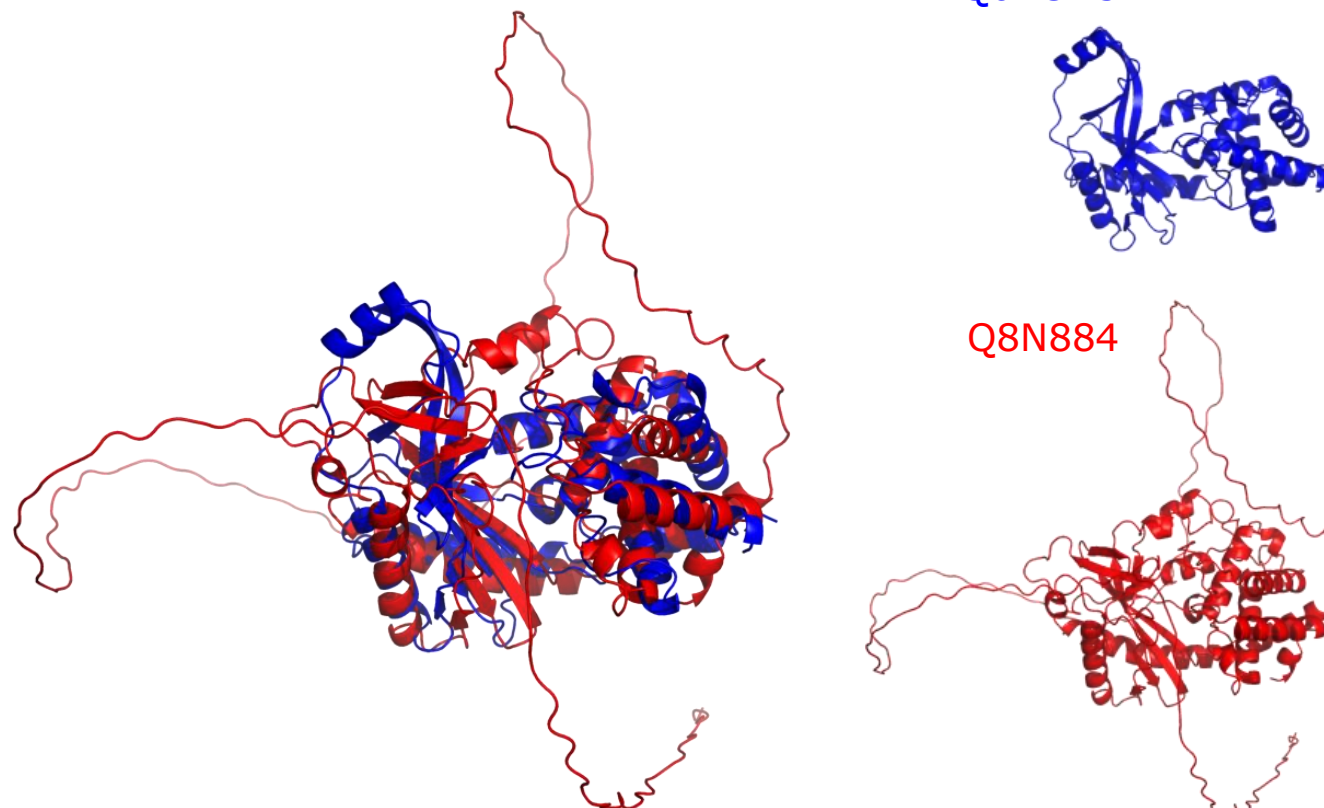
- confers resistance to phage P1
- represses T7 RNA-polymerase action

Q8WXG1 [H. sapiens][RSAD2]

- interferon-inducible antiviral protein
- initiates pathway to repress RNA polymerase action



System	CBASS
PyMol RMSD	5.53 Å
Aln length	1385 atoms [73 AAs]

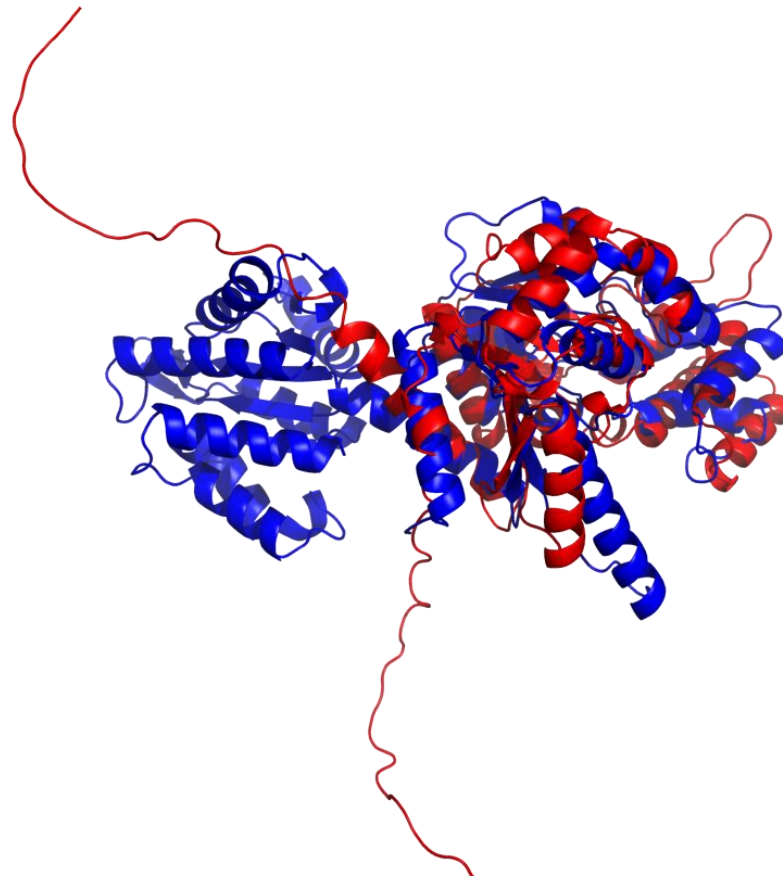
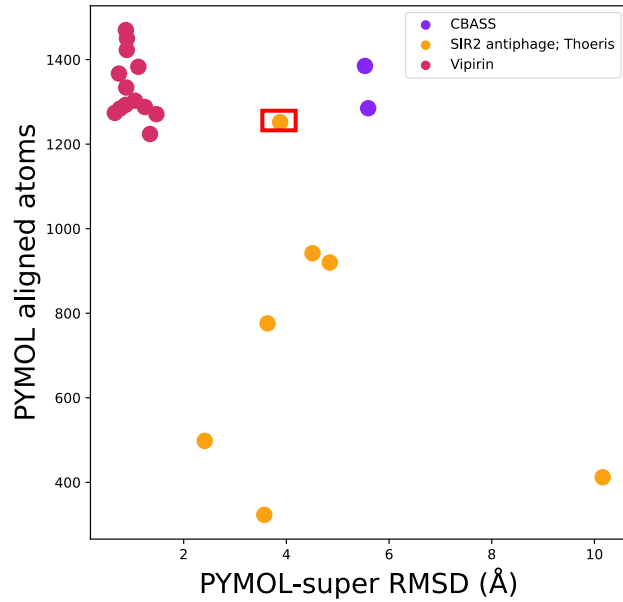


Q6XGD5 [E. coli][cdnE]

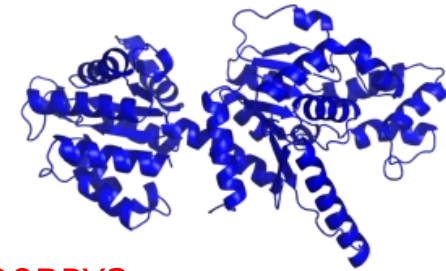
- Cyclic nucleotide synthase (second messenger synthesis) in CBASS system

Q8N884 [H. sapiens][CGAS]

- nucleotidyltransferase catalyzing formation of cyclic GMP-AMP; plays role in innate immunity



J8G6Z1



Q9BPY3



System	SIR2 antiphage
PyMol RMSD	3.88 Å
Aln length	1252 atoms [66 AAs]

J8G6Z1 [*B. cereus*][thsA]

- NAD⁺ hydrolyzing component (NADase) of Thoeris antiviral defense system

Q9BPY3 [*H. sapiens*][FAM118B]

- Cajal body regulation

Next steps

1. Work with V Furlan to further develop composite score for protein pair prioritization
2. Use V Furlan's pipeline frontend to conduct prioritization
 - develop Blastp canonical protein flagger tool and integrate with frontend
3. Work with V Furlan to get a gene list by next meeting (1 - 2 months)
 - target users: C Marshall clinical research group
 - purpose: improve diagnostic yield for rare single-variant immune disease from genome sequencing